

Michael Cusack-Nelkin

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EDUCATION

Columbia University <i>MA in Mathematics of Finance</i>	New York, NY 09/2025 – 12/2026
Coursework: Statistical Aspects of Finance (Dr. Jeremy Shen), Algorithmic Trading, Probability (PhD), Quantitative Portfolio Management	
The College of William & Mary <i>BS in Physics, Minors in Data Science and Mathematics</i>	Williamsburg, VA 09/2018 – 06/2022
- GPA: 3.72 / 4.0 - Coursework: Advanced Applied Machine Learning, Quantum Mechanics I & II, Fluid Mechanics - Honors: Dean's List, Magna Cum Laude	

WORK EXPERIENCE

Deloitte Consulting <i>Deloitte Quantum AI Analyst</i>	Austin, TX 01/2023 – 12/2024
Client Pitching – Pitched the firm's quantum computation and optimization capabilities in presentations to commercial clients from the finance and logistics sectors. Quantum Optimization – Built and presented Python demonstrations of Grover's quantum search algorithm using IBM Qiskit in Python and IBM simulators and quantum hardware.	
Business Technology Analyst, AI and Data	10/2022– 06/2025
Tech Infrastructure – Built modern web infrastructure for the Texas Office of the Inspector General, allowing them to more effectively manage, report, and archive \$140,000,000 of annual Medicaid fraud recoveries. Team Leadership – Led a team of four software developers and testers and shipped a case processing system for Texas OIG to recover \$10,000,000 of previously unrecoverable fraudulent claims upon deployment in 2023.	
Gilder Gagnon & Howe Co. <i>Securities Analysis Intern</i>	New York, NY 02/2018 – 05/2018
Fundamental Analysis – Assisted with fundamental analysis of publicly listed US businesses in the gaming industry. Analyzed 10K/10Q filings and core financial statements and summarized for equity analysts. Stakeholder Participation – Participated in investment committee meetings with portfolio company executives across the media, aggrotech, and biotech industries. Questioned executives' pitches and companies' financials. Took comprehensive notes for equity analysts and company management.	

PERSONAL / ACADEMIC PROJECTS

Statistical Aspects of Finance <i>Optimal Order Splitting with Latency, Replenishment, and OBI</i>	New York, NY 01/2026 – Present
Execution Strategy – Built an optimal execution strategy using L2 order book data. The strategy splits a large parent order into variable quantity child orders and executes them as IOC limit orders based on orderbook imbalance, latency, and book replenishment rate. Built a simulator to test the strategy, which takes into account price impact, Poisson book replenishment, latency, and simulated fills.	
Millennium x Columbia University Trading Competition <i>ETF Arbitrage Simulation</i>	New York, NY 08/2025
Leadership – Led a team of four at an in-person ETF arbitrage competition. Utilized the z-score of historical price residuals to discover trading opportunities. Achieved 1st place in risk-adjusted returns.	
Personal Research Project <i>Exploration of Hidden Markov Models</i>	New York, NY 05/2025 – 06/2025
Learning via Implementation – Explored the mechanics of HMMs by building models from scratch in Python and implementing the Baum-Welch algorithm and state and transition matrices. Successfully modeled binary classification of vowels and consonants using a train/test split over the Brown Corpus as a dataset.	

SKILLS & ADDITIONAL INFORMATION

- Programming:** Python Scientific Stack (Intermediate), SQL (Basic), C++ (Basic), Rust (Basic)
- Interests and Activities:** Alpha Epsilon Pi Jewish Fraternity, Astronomy, Formula 1, Rock Climbing